

Sebastian Bergemann

Researcher • Software Engineer • PhD Candidate

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PROFILE

Researcher and software engineer working at the intersection of software engineering and complex systems engineering. My work focuses on model-based systems engineering, multi-model consistency checking, and confidentiality-preserving computation, with an increasing interest in applying generative AI to engineering tasks. I develop prototypes and software tools that translate research results into practical solutions for real-world engineering applications.

RESEARCH & PROFESSIONAL EXPERIENCE

Researcher

Apr 2021 – Present

fortiss GmbH, Munich, Germany

Research and software development in **model-based systems engineering (MBSE)**, focusing on multi-model consistency, traceability, automated analysis, and engineering automation. **Project Lead** for DigiSemNet (semantic linking of heterogeneous engineering models and GenAI) and **lead researcher/developer** for BaSys4Transfer (automated model consistency checking).

Researching **confidentiality-preserving consistency checking** by investigating and adapting approaches including trusted execution environments, secure multi-party computation, and homomorphic encryption to enable consistency analysis without exposing sensitive information from engineering models.

Translating research concepts into practical engineering tools and demonstrators, including emerging work on **generative AI for semantic relation discovery and engineering automation**, from conceptual design through implementation and evaluation. **Supervised** four student theses, three working students, and five practical courses in MBSE (research transfer).

Research Assistant (Working Student)

Jul 2020 – Dec 2020

fortiss GmbH, Munich, Germany

Developed and evaluated cooperative adaptive cruise control (CACC) functionality for the fortiss Mobility Lab's small-scale autonomous vehicles (MBSE demonstrators), including hardware and system configuration.

Implemented a Python-based orchestration module for skill-based automation systems, automatically generating IEC 61499 control logic from skill interfaces and production sequences using Eclipse 4diac, including support for fault-tolerant execution.

Mechanical Engineer (Dual Study Program)

Oct 2012 – Sep 2015

KSB AG, Frankenthal, Germany

Analyzed and designed an ultrasonic sensor-equipped flow control valve and investigated heat exchanger concepts for specialized pump-engineering applications, culminating in an automated heat exchanger design script.

EDUCATION

Ph.D. in Computer Science

Jul 2022 – Present

Technical University of Munich (TUM), Munich, Germany

Supervisor: Prof. Dr. Alexander Pretschner

Research focus: heterogeneous multi-model consistency checking and confidentiality preservation in systems engineering.

Planned defense: summer 2027.

M.Sc. Robotics, Cognition, Intelligence

Oct 2018 – Mar 2021

Technical University of Munich (TUM), Munich, Germany

Master's thesis: *Evaluation Framework for a Self-Adaptive Cyber-Physical System on Example of a Multi-Robot System.*

Final grade: 1,3 (1="very good", 2="good", ...)

B.Sc. Computer Science

Oct 2015 – Sep 2018

Technical University of Munich (TUM), Munich, Germany

Bachelor's thesis at fortiss: *Architectural Impacts of Introducing Cooperative Adaptive Cruise Control according to ISO/DIS 20035.*

Final grade: 1,9 (1="very good", 2="good", ...)

B.Eng. Mechanical Engineering

Oct 2012 – Sep 2015

DHBW Mannheim, Germany

Dual study program with KSB AG, Frankenthal.

Bachelor's thesis at KSB AG: *Auslegung eines Wärmeaustauschers für die Sonderanwendungen in der Pumpenkonstruktion.*

Final grade: 2,0 (1="very good", 2="good", ...)

INTERNATIONAL EXPERIENCE

Erasmus Exchange (part of the TUM M.Sc. Robotics)

Aug 2019 – May 2020

Royal Institute of Technology (KTH), Stockholm, Sweden

Coursework and practical projects in machine learning, data mining, large-scale development, and multimodal interaction.

RESEARCH INTERESTS

Model-Based Systems Engineering: model-based development and analysis of complex systems, heterogeneous models, model consistency, traceability, and engineering automation.

Software Engineering & AI: methods, tools, and technologies for reliable and maintainable software, including the use of generative AI and LLMs for engineering automation and knowledge integration.

Secure & Confidential Systems: confidentiality-preserving software and systems, trusted execution environments, and secure processing of sensitive engineering data, especially in the field of MBSE.

SELECTED RESEARCH AND INDUSTRY PROJECTS

DigiSemNet <i>BMW-funded research project</i>	Nov 2025 – Present <i>fortiss GmbH</i>
Project Lead, leading a team of three and coordinating partner communication, requirements, and project deliverables. Researching semantic graphs for linking heterogeneous engineering models and integrating LLM- and MCP-based capabilities for automated relation discovery and consistency checking. Developing the integration of consistency checking services into the DigiSemNet platform for use by AI agents.	
Mobility Lab <i>Internal research and demonstrator platform</i>	Aug 2021 – Present <i>fortiss GmbH</i>
Developing and maintaining the Mobility Lab's physical and software demonstrators, including small-scale autonomous vehicles, MBSE tooling, model consistency checking, deployment infrastructure, and simulation components. Supervising working students and leading practical courses, demonstrations, and MBSE trainings showcasing model-based engineering in realistic autonomous-driving scenarios.	
BaSys4Transfer <i>BMFTR-funded research project with industry partners (Festo, Bosch, ABB, Lenze, ZF, ...)</i>	Oct 2022 – Sep 2025 <i>fortiss GmbH</i>
Lead researcher and developer of a work package on model conformity and consistency checking; developed an open-source rule-based consistency checker with an extensible adapter architecture for heterogeneous modeling tools. Designed a flexible OCL-based rule engine and integrated the checker with the BaSyx SDK and an external semantic matching service, while researching confidentiality-preserving consistency checking and publishing the results.	
SpesML <i>BMFTR-funded research project with industry partners (Siemens, Schaeffler, Validas, ...)</i>	Jan 2021 – Mar 2023 <i>fortiss GmbH</i>
Researcher and developer focusing on the technical viewpoint of the SPES methodology; co-designed the SpesML technical architecture model covering hardware, software tasks, and deployment, and implemented it in the SpesML MagicDraw plugin. Designed and implemented traceability mechanisms between logical and technical viewpoints and well-formedness checks, validated through an automotive window-lifter example and contributing to the project's published book on SpesML.	

SELECTED PUBLICATIONS

S. Bergemann, A. Bayha, D. Zhu, M. Sadeghi, C. Atkinson, A. Pretschner, **“Mind the Leak: Formalizing Confidentiality Preservation Assessment of Multi-Model Consistency Checking Systems”**, in SAM/MODELS-C 2025.

S. Bergemann, N. Benkendorf, **“Modular Consistency Checking Between Heterogeneous Models Without Direct Data Exchange Between Collaborators”**, in MODELS-C 2024.

S. Bergemann, **“Towards Confidentiality in Multi-Model Inconsistency Detection for Systems Engineering”**, in MODELS-C 2023.

S. Bergemann, **“Challenges in Multi-View Model Consistency Management for Systems Engineering”**, in Modellierung 2022.

TECHNICAL SKILLS

Programming: Python, Java, C, C++, JavaScript, Clojure
Systems & Modeling: SysML v1/v2, UML, Modelica, AutoFOCUS 3, MATLAB/Simulink, 4diac, Cameo/MagicDraw
Software Engineering: Git(Hub/Lab), Docker, REST APIs, Spring Framework, Ansible, VS Code, Eclipse, IntelliJ IDEA
Data & AI: NumPy, SciPy, Pandas, scikit-learn, NetworkX, Neo4j, OpenAI API, Anthropic API, MCP, Hermes Agent, OpenViking
Security, Privacy & Platforms: Intel SGX, AMD SEV, Occlum, Gramine, FRESCO (MPC), Linux/Ubuntu, Raspberry Pi
Mechanical Engineering & Robotics: CATIA, Siemens NX, Autodesk Fusion 360, ROS 1/2, Gazebo, CARLA, OpenCV

LANGUAGES

German: Native **English:** Fluent